

Total unimodularity

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Solution of the practice quiz

A matrix is totally unimodular if the determinant of each square submatrix is 0, 1 or -1 . We check whether the matrices A , B , C and D satisfy those conditions.

- Matrix A . There are two entries of the matrix different from 0, 1 or -1 . Those are submatrices of size 1 for which the determinant is not 0, 1 or -1 . Therefore, the matrix A is not totally unimodular.
- Matrix B . Each element of the matrix, that is each square submatrix of size 1, is 0, 1 or -1 . Consider now the square submatrices of size 2:

$$\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1, \quad \det \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = 0, \quad \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$$

As there are only two rows, there are no square submatrices of size 3 or above. As all the determinants are 0, 1 or -1 , B is totally unimodular.

- Matrix C . Each element of the matrix, that is, each square submatrix of size 1, is 0, 1 or -1 . If we consider the submatrix containing the first two columns of C , we have

$$\det \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2.$$

Consequently, C is not totally unimodular.

- Matrix D . Each element of the matrix, that is, each square submatrix of size 1, is 0, 1 or -1 . We enumerate the submatrices of size two and

calculate their determinant:

$$\begin{aligned}\det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= 1, & \det \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} &= 0, & \det \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} &= 0, \\ \det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= 1, & \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= 1, & \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= 1, \\ \det \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} &= 0.\end{aligned}$$

They are all 0, 1 or -1 . Finally, we consider the unique submatrix of size 3, which is D itself. As $\det(D) = 1$, D is totally unimodular.